

A Hypothesis on a Structure–Based Group Description of Fluid Dynamics

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Abstract

This work proposes a research hypothesis concerning the possibility of describing fluid dynamics through temporary groups of particles or fluid elements characterized by similar or dynamically correlated motion.

Traditional descriptions of fluid flow are based on physical quantities such as velocity, pressure, density, viscosity, and their spatial and temporal distributions. The proposed approach considers the same system from a Lagrangian perspective, focusing on the trajectories of individual particles or virtual fluid elements and on the identification of collective structures emerging from their motion.

The central hypothesis is that complex fluid motion may be represented as a system of dynamic groups whose members exhibit similar or interconnected kinematic behavior. These groups are not assumed to remain fixed: they may emerge, evolve, split, merge, or disappear over time.

Particular attention is given to how such structures reorganize under controlled perturbations, how perturbations propagate through the fluid, how the response changes with depth, and whether subsequent system dynamics may depend on the history of preceding perturbations.

This work does not claim in advance that the proposed approach is fundamentally new or capable of replacing existing equations of fluid dynamics. It is presented as a research hypothesis and a computational research program aimed at identifying reproducible patterns in collective fluid dynamics.

Keywords: fluid dynamics, Lagrangian approach, particle trajectories, collective motion, coherent structures, turbulence, perturbations, trajectory clustering, transport, dynamic structures.

1. Introduction

Fluid motion is a complex dynamical process in which a large number of interacting elements produce observable macroscopic flow.

At the macroscopic level, a fluid can be described as a continuous medium through fields such as velocity, pressure, density, and other physical quantities.

However, complex flow can also be investigated through the trajectories of individual particles or virtual fluid elements.

This leads to an additional research question:

Can complex fluid motion be described not only through the instantaneous state of a field, but also through temporary collective structures of motion evolving over time?

The present work focuses on this question.

2. Main Hypothesis

It is hypothesized that complex flow may contain temporary groups of particles characterized by similar or dynamically correlated trajectories.

Such a group is not necessarily a fixed collection of particles. Its membership may change over time.

Therefore, a group is considered here as a dynamic kinematic structure defined primarily by the character of motion rather than by permanent particle membership.

Conceptually, the flow may be represented as:

**individual elements → groups of similar motion → interaction between groups
→ restructuring of collective dynamics.**

3. Hypothesis Concerning the Transition from Individual to Collective Motion

The proposed research investigates how the character of motion changes as the number of interacting elements increases.

With a small number of elements, individual trajectories may be directly observed and analyzed.

As the number of elements increases, collective behavior becomes more complex.

The following hypothesis is proposed:

As the number of interacting elements increases, a description based on individual trajectories may become less informative, while a description based on collective dynamic structures may become more informative.

This hypothesis requires computational and experimental verification.

4. Identification of Dynamic Groups

Particles may be classified according to similarities in their motion.

For example, a flow may contain:

- Group A — particles predominantly moving in one direction;
- Group B — particles exhibiting another dominant direction of motion;
- Group C — particles participating in rotational or vortex-like motion;
- Group D — particles located in transitional regions between different types of motion.

The groups should preferably be identified using measurable characteristics rather than visual inspection alone.

Potential characteristics include:

- position;

- velocity;
 - direction;
 - acceleration;
 - trajectory curvature;
 - local vorticity;
 - trajectory similarity;
 - temporal correlation of motion.
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5. Dynamics of Groups

The proposed concept assumes that dynamic groups may have a life cycle:

formation → development → interaction → modification → splitting or merging → disappearance.

Possible measurements include:

- group lifetime;
- change in group size;
- change in shape;
- change in mean velocity;
- change in direction;
- group fragmentation;
- group merging;
- formation of new groups.

Hypothesis H1

Changes in the structure of dynamic groups may provide additional information about the temporal evolution of a flow that is not fully represented by instantaneous macroscopic parameters.

6. Perturbation as a Research Tool

Controlled external perturbations can be used to investigate system dynamics.

Possible perturbations include:

1. horizontal forcing from left to right;
2. horizontal forcing from right to left;
3. localized forcing;
4. surface forcing;
5. short-duration impulses;
6. periodic forcing;
7. forcing with different intensities.

Following each perturbation, changes in particle trajectories and dynamic groups can be tracked.

The central question is:

How does the collective structure of motion change following a controlled perturbation?

7. Propagation of Perturbations

A localized perturbation may produce changes in other regions of the fluid.

The proposed approach is to investigate this propagation through the sequence of reorganizations of dynamic groups.

The following quantities may be examined:

- location of the initial response;
- groups responding directly;
- groups responding with a delay;
- propagation rate;
- groups that persist;
- groups that break apart;
- newly formed groups.

Hypothesis H2

A localized external perturbation may cause a sequential reorganization of neighboring dynamic structures, resulting in a larger-scale restructuring of the flow.

8. Depth-Dependent Response

A separate direction of investigation concerns perturbations applied at the fluid surface.

The fluid may be divided into several depth levels:

- surface layer;
- intermediate layer;
- deeper layer.

Following a surface perturbation, particle motion can be monitored at each depth.

The analysis may include:

- response time;
- response intensity;
- directional changes;
- velocity changes;
- formation or disappearance of dynamic groups;
- restructuring of motion.

Hypothesis H3

The response of a fluid to the same surface perturbation may systematically depend on depth.

9. Comparison of Different Perturbation Types

Several forcing regimes can be compared.

9.1. Relatively Calm State

The initial organization of the fluid is examined without a deliberately imposed perturbation.

9.2. Horizontal Forcing

A directed perturbation is applied and responses to different forcing directions are compared.

9.3. Localized Forcing

The perturbation is restricted to a limited spatial region. Its subsequent propagation is investigated.

9.4. Surface Forcing

A perturbation is applied at the surface, followed by analysis of the response at different depths.

9.5. Periodic Forcing

The perturbation is repeated at a defined frequency. The possible formation of persistent or recurring structures is investigated.

10. Possible Dependence on System History

The proposed research examines whether subsequent fluid evolution depends only on its instantaneous state or may also depend on preceding perturbations.

Two states with similar instantaneous characteristics may be produced through different forcing histories.

For example:

Scenario A: calm state → perturbation 1 → current state.

Scenario B: perturbation 2 → evolution → current state.

Their subsequent evolution can then be compared.

Hypothesis H4

Fluid states with similar instantaneous parameters may exhibit different subsequent dynamics depending on their preceding forcing history.

11. Geometric Representation of Dynamic Structures

The possibility of describing dynamic groups geometrically may also be investigated.

For each dynamic structure, potential descriptors include:

- spatial position;
- size;
- shape;
- direction of motion;
- mean velocity;
- velocity distribution;
- rotational character;
- connectivity;
- lifetime;
- rate of structural change.

Complex flow could therefore be represented as a collection of time-dependent structures:

$G_1(t), G_2(t), G_3(t), \dots, G_n(t)$.

Importantly, $G_i(t)$ does not necessarily contain the same particles throughout its lifetime.

12. Possible Mathematical Description

At the initial stage, the research does not propose a new fundamental equation.

The first objective is to determine whether reproducible patterns exist.

If such patterns are identified, mathematical descriptions may then be developed for:

- group formation;
- group destruction;
- merging;
- splitting;
- transitions between groups;
- group lifetime;
- dependence on external forcing;
- dependence on depth;
- dependence on spatial scale.

One possible abstract representation is:

$$dG_i/dt = F(G_i, G_j, \dots, E, B),$$

where G_i represents a dynamic structure, E represents external forcing parameters, and B represents characteristics of the surrounding environment.

This is not proposed as a finished physical law. It represents only a possible direction for future mathematical research.

13. Computational Experiment

The initial investigation can be performed using computer simulation.

A numerical fluid model capable of tracking virtual particles or Lagrangian elements can be used.

For each element, the following quantities may be recorded:

- coordinates;
- velocity;

- acceleration;
- direction of motion;
- trajectory;
- time.

Trajectory similarity can then be analyzed to identify dynamic groups.

The main experimental cycle is:

initial state → motion recording → grouping → perturbation → continued recording → regrouping → comparison.

14. Before–During–After Analysis

A particularly important aspect of the proposed methodology is comparison across three temporal states.

Before the perturbation — the initial structure is determined.

During the perturbation — the immediate response is recorded.

After the perturbation — subsequent restructuring is analyzed.

This makes it possible to study not only the state of the system but also the transition between states.

15. Conceptual Model of “Interwoven Threads”

Complex flow may be conceptually represented as a large number of differently colored threads intertwined with one another.

In this analogy, each thread represents a dynamic structure of motion.

After a perturbation, structures may:

- bend;
- stretch;

- split;
- merge;
- rotate;
- disappear;
- form new structures.

This analogy is not intended as a physical description of a fluid. It is a conceptual model for asking whether apparently chaotic motion contains an underlying organization of interacting collective structures.

The central question is:

Can a seemingly chaotic flow be decomposed into a system of identifiable, interacting dynamic structures?

16. Main Research Question

The proposed research can be summarized by the following question:

Can complex fluid dynamics be described through the temporal structure of collective kinematic groups and the regularities governing their reorganization under external perturbations?

Additional questions include:

1. How many dynamic structures coexist at a given time?
2. How long do they persist?
3. Which characteristics determine their stability?
4. How do the structures interact?
5. How do splitting and merging occur?
6. How does a perturbation propagate between structures?
7. How does the response depend on depth?
8. Does subsequent evolution depend on forcing history?
9. Can universal characteristics of such structures be identified?
10. Can a mathematical model of their formation and interaction be constructed?

17. Potential Scientific Significance

The proposed approach is not intended to replace existing equations of fluid dynamics.

Its purpose is to investigate whether an additional descriptive level can be identified:

not only describing the instantaneous state of a fluid field, but also identifying collective structures of motion, their formation history, and the regularities governing their reorganization over time.

If such structures demonstrate stable and reproducible characteristics, they may potentially provide useful descriptors for complex flows, material transport, mixing, propagation of perturbations, and other dynamical processes.

18. Limitations

Several limitations should be explicitly acknowledged.

First, trajectory grouping and the study of collective structures are already present in various areas of modern fluid dynamics and flow analysis. Therefore, this work does not claim that the basic idea of grouping is fundamentally new.

Second, the existence of related methods does not exclude the possibility that new combinations of grouping criteria, forcing regimes, and analytical procedures may reveal additional patterns.

Third, a numerical simulation does not by itself establish the universality of a result.

Fourth, a pattern observed in one model must be tested under different parameters, scales, boundary conditions, and physical regimes.

Accordingly, the proposed project should be regarded as a research hypothesis and a program for systematic investigation.

19. Conclusion

This work proposes a conceptual framework for investigating fluids as dynamic systems in which large numbers of elements may temporarily organize into collective structures of motion.

The proposed approach focuses not only on individual trajectories but also on dynamic groups emerging from those trajectories, their life cycles, and their interactions.

Particular attention is given to controlled perturbations and transitions between states:

calm state → perturbation → change in motion → restructuring of groups → new state.

A separate research direction concerns the propagation of surface perturbations into deeper layers of the fluid and comparison of responses at different depths.

The initial testing of these hypotheses can be performed through computer simulation and tracking of virtual particles.

If stable and reproducible patterns are identified, the next stage would be their formalization, mathematical modeling, and comparison with established methods for describing fluid flows.

Thus, the present work is proposed as a research program aimed at investigating whether an additional structural level of fluid dynamics can be described through collective kinematic structures and their temporal reorganization.